

Kristýna Harvanová

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🌐 Kristyna-Harvanova
in Kristýna Harvanová

Education

- 2024–present **Master's Degree, Charles University, Faculty of Mathematics and Physics, Institute of Formal and Applied Linguistics**
Major: Computer Science - Language Technologies and Computational Linguistics
- 2021–2024 **Bachelor's Degree, Charles University, Faculty of Mathematics and Physics**
Major: Computer Science, specialization: Artificial Intelligence
- 2025 **Micro-credential certificate CU - Deep Learning**
- 2025–present **Student of a programm PRG.AI Minor**
- 2021–2024 **Jaroslav Ježek's Conservatory and College of Music**, major: Music, specialization: piano, piano - jazz

Technical Skills

- Languages Python, Julia, Java, C#, non-procedural languages, TypeScript, $\LaTeX$
- ML & Data PyTorch, scikit-learn, OpenCV, HuggingFace Transformers, pandas, NumPy, Dash
- Tools & Infra AWS (EC2, S3), Databricks, MLflow, Docker, Git, Dash, React Native, Expo, Linux CLI

Work Experience

- 2024–present **Data Scientist Intern, MSD (Merck)**, member of a global Data Science & Scientific Informatics team.
 - Image segmentation for pharmaceutical research. Video analysis in life sciences
 - NLP for enterprise search – tokenization, query processing, evidence highlighting
 - Designed internal tooling for reproducible AI experiments (Claude Code)
 - Built interactive dashboards and demo applications for domain researchers
- 2025–present **Teacher: Information and Communication Technology, Obchodní akademie a lyceum Heroldovy Sady**
- 2023–2025 **Member of Prague Music Computing Group, Institute of Formal and Applied Linguistics**

Projects

- 2024 **Optical Music Recognition – BSc Thesis, Python, YOLO, OpenCV, synthetic data augmentation**
Postprocessing of Synthetic Sheet Music in OMR. Improved symbol detection mAP to 0.954
- 2023 **Other projects**
 - C# PacMan – implementation of search algorithms (e.g., A*) in C#, GUI
 - Java Jazz Chord Generator – OOP, generic types, command line app, unit testing
 - React Native Parking Reminder – cross-platform mobile parking app (TypeScript, Expo, geofencing, local notifications)
 - Prolog Rubik's Cube Solver – using non-procedural and logic programming

Achievements

- 2024 **Active contribution at the Prague Music Computing Group workshop**
"Postprocessing of Synthetic Sheet Music in the Context of Optical Music Recognition"
- 2022, 23, 25 **Academic excellence scholarship, MFF UK**
- Competitions **Logic Olympiad, Mathematical Olympiad, B. Challenge (national), Czechitas IT (1st place)**

Language and Other Skills

- Czech Native speaker
- English C1, Cambridge Assessment English FCE (level B2) from 2019
- German B2, ÖSD Zertifikat B2 (level B2)
- Driving licence Category B

Interests

- Music Jazz conservatory student, member of professional Kühn's Choir, drums, piano&music teacher
- Dance & Sport Musical, Latin dance, Badminton, swimming, yoga, bouldering